LibJuno Requirements Traceability Matrix (RTM)

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1. Coverage Summary

Metric	Count	Percentage
Total requirements	188	—
Traced to code	188	100%
Traced to design	188	100%
Traced to tests (of test-verified)	121	100%

2. Per-Module Summary

Module	Requirements	Traced to Code	Traced to Design	Traced to Tests
APP	4	4	4	0
ARRAY	7	7	7	7
CRC	10	10	10	10
HASH	3	3	3	2
HEAP	8	8	8	8
IO	14	14	14	0
LOG	7	7	7	1
MAP	10	10	10	10
MATH	18	18	18	14
MEMORY	10	10	10	8
MODULE	6	6	6	0
MTX	5	5	5	1
POINTER	5	5	5	4
QUEUE	8	8	8	8
SB	13	13	13	9
SCH	7	7	7	3
SM	11	11	11	8
STACK	8	8	8	8
STATUS	4	4	4	0
SYS	11	11	11	3
TIME	19	19	19	17

3. Traceability Matrix

3.1. Module: APP

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-APP-001: Application Root Structure	include/juno/app/app_api.h:52 (JUNO_MODULE_ROOT)	docs/sdd/modules/application.adoc:1 (Application Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-APP-002: OnStart Lifecycle Hook	include/juno/app/app_api.h:58 (JUNO_STATUS_T)	docs/sdd/modules/application.adoc:1 (Application Module)	*none*	Inspection	Complete
REQ-APP-003: OnProcess Lifecycle Hook	include/juno/app/app_api.h:61 (JUNO_STATUS_T)	docs/sdd/modules/application.adoc:1 (Application Module)	*none*	Inspection	Complete
REQ-APP-004: OnExit Lifecycle Hook	include/juno/app/app_api.h:64 (JUNO_STATUS_T)	docs/sdd/modules/application.adoc:1 (Application Module)	*none*	Inspection	Complete

3.2. Module: ARRAY

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-ARRAY-001: Array Initialization	include/juno/ds/array_api.h:163 (JunoDs_ArrayInit)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:150 (test_array_init_success) tests/test_array.c:162 (test_array_init_null_array)	Test	Complete
REQ-ARRAY-002: Array SetAt	include/juno/ds/array_api.h:82 (JUNO_STATUS_T)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:174 (test_array_set_at)	Test	Complete
REQ-ARRAY-003: Array GetAt	include/juno/ds/array_api.h:89 (JUNO_RESULT_POINTER_T)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:184 (test_array_get_at)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-ARRAY-004: Array RemoveAt	include/juno/ds/array_api.h:96 (JUNO_STATUS_T)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:195 (test_array_remove_at)	Test	Complete
REQ-ARRAY-005: Array Bounds Checking	include/juno/ds/array_api.h:140 (JunoDs_ArrayVerifyIndex)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:208 (test_array_set_at_oob) tests/test_array.c:218 (test_array_get_at_oob) tests/test_array.c:226 (test_array_remove_at_oob) tests/test_queue.c:913 (test_queue_array_api_get_at_out_of_bounds) tests/test_queue.c:923 (test_queue_array_api_remove_at_out_of_bounds) tests/test_stack.c:926 (test_stack_array_api_get_at_out_of_bounds) tests/test_stack.c:936 (test_stack_array_api_remove_at_out_of_bounds)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-ARRAY-006: Array Vtable Validation	include/juno/ds/array_api.h:120 (JunoDs_ArrayVerify)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:238 (test_array_verify_null_api) tests/test_queue.c:892 (test_queue_array_api_set_at_invalid_api) tests/test_stack.c:905 (test_stack_array_api_set_at_invalid_api) tests/test_stack.c:946 (test_stack_array_api_null_array)	Test	Complete
REQ-ARRAY-007: Array Capacity Validation	include/juno/ds/array_api.h:120 (JunoDs_ArrayVerify) include/juno/ds/array_api.h:163 (JunoDs_ArrayInit)	docs/sdd/modules/array.adoc:1 (Array Module)	tests/test_array.c:253 (test_array_verify_zero_capacity) tests/test_array.c:264 (test_array_init_zero_capacity)	Test	Complete

3.3. Module: CRC

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-CRC-001: CRC Streaming Update	src/juno_arc.c:21 (Juno_CrcArcUpdate) src/juno_binhex.c:21 (Juno_CrcBinhexUpdate) src/juno_ccitt.c:21 (Juno_CrcCcittUpdate) src/juno_ccitt32.c:20 (Juno_CrcCcitt32Update) src/juno_kermit.c:21 (Juno_CrcKermitUpdate) src/juno_zip.c:20 (Juno_CrcZipUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:81 (test_various_crc_behaviour)	Test	Complete
REQ-CRC-002: CRC Null Safety	src/juno_arc.c:21 (Juno_CrcArcUpdate) src/juno_binhex.c:21 (Juno_CrcBinhexUpdate) src/juno_ccitt.c:21 (Juno_CrcCcittUpdate) src/juno_ccitt32.c:20 (Juno_CrcCcitt32Update) src/juno_kermit.c:21 (Juno_CrcKermitUpdate) src/juno_zip.c:20 (Juno_CrcZipUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:70 (test_empty_crc_all)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-CRC-003: CRC Zero Length Safety	src/juno_arc.c:21 (Juno_CrcArcUpdate) src/juno_binhex.c:21 (Juno_CrcBinhexUpdate) src/juno_ccitt.c:21 (Juno_CrcCcittUpdate) src/juno_ccitt32.c:20 (Juno_CrcCcitt32Update) src/juno_kermit.c:21 (Juno_CrcKermitUpdate) src/juno_zip.c:20 (Juno_CrcZipUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:70 (test_empty_crc_all)	Test	Complete
REQ-CRC-004: CRC-16 ARC	src/juno_arc.c:21 (Juno_CrcArcUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:81 (test_various_crc_behaviour)	Test	Complete
REQ-CRC-005: CRC-16 BinHex	src/juno_binhex.c:21 (Juno_CrcBinhexUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:81 (test_various_crc_behaviour)	Test	Complete
REQ-CRC-006: CRC-16 CCITT	src/juno_ccitt.c:21 (Juno_CrcCcittUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:56 (test_ccitt_crc)	Test	Complete
REQ-CRC-007: CRC-16 Kermit	src/juno_kermit.c:21 (Juno_CrcKermitUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:81 (test_various_crc_behaviour)	Test	Complete
REQ-CRC-008: CRC-32 CCITT32	src/juno_ccitt32.c:20 (Juno_CrcCcitt32Update)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:81 (test_various_crc_behaviour)	Test	Complete
REQ-CRC-009: CRC-32 ZIP	src/juno_zip.c:20 (Juno_CrcZipUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:37 (test_zip_crc)	Test	Complete
REQ-CRC-010: CRC-32 ZIP Final XOR	src/juno_zip.c:20 (Juno_CrcZipUpdate)	docs/sdd/modules/crc.adoc:1 (CRC Module)	tests/test_crc.c:37 (test_zip_crc)	Test	Complete

3.4. Module: HASH

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HASH-001: DJB2 Hash Computation	include/juno/hash/hash_djb2.h:47 (JunoHash_Djb2)	docs/sdd/modules/hash.adoc:1 (Hash Module)	tests/test_hash.c:37 (test_hash_djb2_known_value) tests/test_hash.c:52 (test_hash_djb2_empty_buffer) tests/test_hash.c:62 (test_hash_djb2_different_inputs_differ)	Test	Complete
REQ-HASH-002: DJB2 Null Buffer Rejection	include/juno/hash/hash_djb2.h:47 (JunoHash_Djb2)	docs/sdd/modules/hash.adoc:1 (Hash Module)	tests/test_hash.c:78 (test_hash_djb2_null_buffer)	Test	Complete
REQ-HASH-003: DJB2 Result Type	include/juno/hash/hash_djb2.h:47 (JunoHash_Djb2)	docs/sdd/modules/hash.adoc:1 (Hash Module)	*none*	Inspection	Complete

3.5. Module: HEAP

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HEAP-001: Heap Initialization	src/juno_heap.c:32 (JunoDs_Heap_Init)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:353 (test_heap_init_nominal) tests/test_heap.c:363 (test_heap_init_null_heap) tests/test_heap.c:373 (test_heap_init_null_api) tests/test_heap.c:383 (test_heap_init_zero_capacity)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HEAP-002: Heap Element Comparison Injection	src/juno_heap.c:32 (JunoDs_Heap_Init)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:526 (test_heap_min_heap_insert_and_property) tests/test_heap.c:709 (test_heap_update_with_compare_error) tests/test_heap.c:727 (test_heap_update_with_swap_error) tests/test_heap.c:825 (test_heap_siftdown_with_compare_error) tests/test_heap.c:844 (test_heap_siftdown_with_swap_error) tests/test_heap.c:909 (test_heap_compare_function_error) tests/test_heap.c:931 (test_heap_swap_function_error) tests/test_heap.c:994 (test_heap_mixed_min_max_operations)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HEAP-003: Heap Insert	src/juno_heap.c:258 (JunoDs_Heap_Insert)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:415 (test_heap_insert_single_item_max_heap) tests/test_heap.c:432 (test_heap_insert_multiple_items_max_heap) tests/test_heap.c:453 (test_heap_insert_to_full_capacity) tests/test_heap.c:472 (test_heap_insert_overflow) tests/test_heap.c:495 (test_heap_insert_null_heap) tests/test_heap.c:505 (test_heap_insert_invalid_pointer) tests/test_heap.c:526 (test_heap_min_heap_insert_and_property) tests/test_heap.c:653 (test_heap_update_single_element) tests/test_heap.c:669 (test_heap_update_bubble_up) tests/test_heap.c:691 (test_heap_update_empty_heap) tests/test_heap.c:702 (test_heap_update_null_heap) tests/test_heap.c:957 (test_heap_stress_insert_pop_cycles) tests/test_heap.c:994 (test_heap_mixed_min_max_operations) tests/test_heap.c:1033 (test_heap_single_element_capacity) tests/test_heap.c:1058 (test_heap_data_into	Test	▯ Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HEAP-004: Heap Pop	src/juno_heap.c:325 (JunoDs_Heap_Pop)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:550 (test_heap_pop_single_item_max_heap) tests/test_heap.c:574 (test_heap_pop_multiple_items_max_heap) tests/test_heap.c:610 (test_heap_pop_empty_heap) tests/test_heap.c:622 (test_heap_pop_null_heap) tests/test_heap.c:631 (test_heap_pop_invalid_pointer) tests/test_heap.c:749 (test_heap_siftdown_from_root) tests/test_heap.c:767 (test_heap_siftdown_from_middle) tests/test_heap.c:787 (test_heap_siftdown_leaf_node) tests/test_heap.c:807 (test_heap_siftdown_empty_heap) tests/test_heap.c:818 (test_heap_siftdown_null_heap) tests/test_heap.c:957 (test_heap_stress_insert_pop_cycles) tests/test_heap.c:994 (test_heap_mixed_min_max_operations) tests/test_heap.c:1033 (test_heap_single_element_capacity) tests/test_heap.c:1058 (test_heap_data_integrity_after_operations)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-HEAP-005: Heap Pop Output	src/juno_heap.c:325 (JunoDs_Heap_Pop)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:550 (test_heap_pop_single_item_max_heap) tests/test_heap.c:574 (test_heap_pop_multiple_items_max_heap) tests/test_heap.c:1058 (test_heap_data_integrity_after_operations)	Test	Complete
REQ-HEAP-006: Heap Heapify	src/juno_heap.c:284 (JunoDs_Heap_Heapify)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:867 (test_heap_heapify_from_unsorted_array) tests/test_heap.c:886 (test_heap_heapify_empty_heap) tests/test_heap.c:898 (test_heap_heapify_null_heap)	Test	Complete
REQ-HEAP-007: Heap Vtable Validation	include/juno/ds/heap_api.h:227 (JunoDs_Heap_Verify)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:397 (test_heap_verify_null_heap) tests/test_heap.c:403 (test_heap_verify_invalid_api)	Test	Complete
REQ-HEAP-008: Heap Dependency Validation	include/juno/ds/heap_api.h:227 (JunoDs_Heap_Verify)	docs/sdd/modules/heap.adoc:1 (Heap Module)	tests/test_heap.c:373 (test_heap_init_null_api) tests/test_heap.c:383 (test_heap_init_zero_capacity)	Test	Complete

3.6. Module: IO

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-IO-001: Async IO Root Structure	include/juno/io/async_io_api.h:47 (JUNO_MODULE_ROOT)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-002: Async IO Read	include/juno/io/async_io_api.h:53 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-003: Async IO TryRead	include/juno/io/async_io_api.h:56 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-004: Async IO ReadUntil	include/juno/io/async_io_api.h:59 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-005: Async IO TryReadUntil	include/juno/io/async_io_api.h:62 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-006: Async IO Write	include/juno/io/async_io_api.h:65 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-007: Async IO TryWrite	include/juno/io/async_io_api.h:68 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-008: Async IO Poll	include/juno/io/async_io_api.h:71 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-009: I2C IO Root Structure	include/juno/io/i2c_io_api.h:108 (JUNO_MODULE_ROOT)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-IO-010: I2C Transfer Operation	include/juno/io/i2c_io_api.h:134 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-011: I2C Message Type Discrimination	include/juno/io/i2c_io_api.h:58 (<unknown>)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-012: I2C Device Address	include/juno/io/i2c_io_api.h:58 (<unknown>)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-013: SPI IO Root Structure	include/juno/io/spi_io_api.h:46 (JUNO_MODULE_ROOT)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete
REQ-IO-014: SPI Transaction Operation	include/juno/io/spi_io_api.h:58 (JUNO_STATUS_T)	docs/sdd/modules/io.adoc:1 (IO Module)	*none*	Inspection	Complete

3.7. Module: LOG

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-LOG-001: Log Root Structure	include/juno/log/log_api.h:51 (JUNO_MODULE_ROOT)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete
REQ-LOG-002: Debug Level Logging	include/juno/log/log_api.h:57 (JUNO_STATUS_T)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-LOG-003: Info Level Logging	include/juno/log/log_api.h:60 (JUNO_STATUS_T)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete
REQ-LOG-004: Warning Level Logging	include/juno/log/log_api.h:63 (JUNO_STATUS_T)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete
REQ-LOG-005: Error Level Logging	include/juno/log/log_api.h:66 (JUNO_STATUS_T)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete
REQ-LOG-006: Log Initialization	include/juno/log/log_api.h:71 (JunoLog_LogInit)	docs/sdd/modules/log.adoc:1 (Log Module)	tests/test_log.c:94 (test_log_init_success) tests/test_log.c:102 (test_log_init_null_root)	Test	Complete
REQ-LOG-007: Variadic Message Format	include/juno/log/log_api.h:57 (JUNO_STATUS_T) include/juno/log/log_api.h:60 (JUNO_STATUS_T) include/juno/log/log_api.h:63 (JUNO_STATUS_T) include/juno/log/log_api.h:66 (JUNO_STATUS_T)	docs/sdd/modules/log.adoc:1 (Log Module)	*none*	Inspection	Complete

3.8. Module: MAP

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MAP-001: Map Initialization	src/juno_map.c:34 (JunoDs_MapInit)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:322 (test_map_init_nominal) tests/test_map.c:336 (test_map_init_null_map) tests/test_map.c:343 (test_map_init_null_api) tests/test_map.c:351 (test_map_init_zero_capacity)	Test	Complete
REQ-MAP-002: Map Hash Function Injection	src/juno_map.c:34 (JunoDs_MapInit)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:322 (test_map_init_nominal) tests/test_map.c:901 (test_map_entry_hash_nominal) tests/test_map.c:912 (test_map_entry_is_null_true) tests/test_map.c:923 (test_map_entry_is_null_false)	Test	Complete
REQ-MAP-003: Map Equality Injection	src/juno_map.c:34 (JunoDs_MapInit)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:322 (test_map_init_nominal) tests/test_map.c:873 (test_map_entry_equals_same_key) tests/test_map.c:887 (test_map_entry_equals_different_key)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MAP-004: Map Set	src/juno_map.c:135 (JunoDs_MapSet)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:365 (test_map_set_single_item) tests/test_map.c:384 (test_map_set_multiple_items) tests/test_map.c:406 (test_map_set_update_existing) tests/test_map.c:430 (test_map_set_with_collision) tests/test_map.c:455 (test_map_set_null_map) tests/test_map.c:663 (test_map_multiple_collisions_linear_probing) tests/test_map.c:714 (test_map_wraparound_probing) tests/test_map.c:741 (test_map_single_element_capacity) tests/test_map.c:776 (test_map_set_get_remove_cycle)	Test	Complete
REQ-MAP-005: Map Table Full Rejection	src/juno_map.c:135 (JunoDs_MapSet)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:465 (test_map_set_table_full)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MAP-006: Map Get	src/juno_map.c:109 (JunoDs_MapGet)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:490 (test_map_get_existing_item) tests/test_map.c:528 (test_map_get_after_collision) tests/test_map.c:556 (test_map_get_null_map) tests/test_map.c:663 (test_map_multiple_collisions_linear_probing) tests/test_map.c:714 (test_map_wraparound_probing) tests/test_map.c:741 (test_map_single_element_capacity) tests/test_map.c:776 (test_map_set_get_remove_cycle)	Test	Complete
REQ-MAP-007: Map Key Not Found	src/juno_map.c:109 (JunoDs_MapGet)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:515 (test_map_get_nonexistent_item)	Test	Complete
REQ-MAP-008: Map Remove	src/juno_map.c:156 (JunoDs_MapRemove)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:570 (test_map_remove_existing_item) tests/test_map.c:593 (test_map_remove_nonexistent_item) tests/test_map.c:620 (test_map_remove_after_collision) tests/test_map.c:649 (test_map_remove_null_map) tests/test_map.c:776 (test_map_set_get_remove_cycle)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MAP-009: Map Vtable Validation	include/juno/ds/map_api.h:130 (JunoDs_MapVerify)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:810 (test_map_verify_valid_map) tests/test_map.c:820 (test_map_verify_null_map)	Test	Complete
REQ-MAP-010: Map Dependency Validation	include/juno/ds/map_api.h:130 (JunoDs_MapVerify)	docs/sdd/modules/map.adoc:1 (Map Module)	tests/test_map.c:343 (test_map_init_null_api) tests/test_map.c:351 (test_map_init_zero_capacity)	Test	Complete

3.9. Module: MATH

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-001: Math Constants	include/juno/math/juno_math_constants.h:33 (Pi)	docs/sdd/modules/math.adoc:1 (Math Module)	*none*	Inspection	Complete
REQ-MATH-002: Vector Types	include/juno/math/juno_vec_types.h:64 (<unknown>)	docs/sdd/modules/math.adoc:1 (Math Module)	*none*	Inspection	Complete
REQ-MATH-003: Vector Coordinate Access	include/juno/math/juno_vec_types.h:64 (<unknown>)	docs/sdd/modules/math.adoc:1 (Math Module)	*none*	Inspection	Complete
REQ-MATH-004: Quaternion Type	include/juno/math/juno_vec_types.h:674 (<unknown>)	docs/sdd/modules/math.adoc:1 (Math Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-005: Vector Addition	include/juno/math/juno_vec.h:67 (Juno_Vec2_F64_Add) include/juno/math/juno_vec.h:139 (Juno_Vec2_F32_Add) include/juno/math/juno_vec.h:201 (Juno_Vec2_I32_Add) include/juno/math/juno_vec.h:264 (Juno_Vec3_F64_Add) include/juno/math/juno_vec.h:335 (Juno_Vec3_F32_Add) include/juno/math/juno_vec.h:405 (Juno_Vec3_I32_Add)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:35 (test_vec2_f64_add) tests/test_math.c:102 (test_vec2_f32_add) tests/test_math.c:169 (test_vec2_i32_add) tests/test_math.c:236 (test_vec3_f64_add) tests/test_math.c:306 (test_vec3_f32_add) tests/test_math.c:376 (test_vec3_i32_add)	Test	Complete
REQ-MATH-006: Vector Subtraction	include/juno/math/juno_vec.h:82 (Juno_Vec2_F64_Sub) include/juno/math/juno_vec.h:151 (Juno_Vec2_F32_Sub) include/juno/math/juno_vec.h:213 (Juno_Vec2_I32_Sub) include/juno/math/juno_vec.h:277 (Juno_Vec3_F64_Sub) include/juno/math/juno_vec.h:348 (Juno_Vec3_F32_Sub) include/juno/math/juno_vec.h:418 (Juno_Vec3_I32_Sub)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:48 (test_vec2_f64_sub) tests/test_math.c:115 (test_vec2_f32_sub) tests/test_math.c:182 (test_vec2_i32_sub) tests/test_math.c:249 (test_vec3_f64_sub) tests/test_math.c:319 (test_vec3_f32_sub) tests/test_math.c:389 (test_vec3_i32_sub)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-007: Vector Scalar Multiplication	include/juno/math/juno_vec.h:96 (Juno_Vec2_F64_Mult) include/juno/math/juno_vec.h:162 (Juno_Vec2_F32_Mult) include/juno/math/juno_vec.h:224 (Juno_Vec2_I32_Mult) include/juno/math/juno_vec.h:289 (Juno_Vec3_F64_Mult) include/juno/math/juno_vec.h:360 (Juno_Vec3_F32_Mult) include/juno/math/juno_vec.h:430 (Juno_Vec3_I32_Mult)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:61 (test_vec2_f64_mult) tests/test_math.c:128 (test_vec2_f32_mult) tests/test_math.c:195 (test_vec2_i32_mult) tests/test_math.c:262 (test_vec3_f64_mult) tests/test_math.c:332 (test_vec3_f32_mult) tests/test_math.c:402 (test_vec3_i32_mult)	Test	Complete
REQ-MATH-008: Vector Dot Product	include/juno/math/juno_vec.h:108 (Juno_Vec2_F64_Dot) include/juno/math/juno_vec.h:173 (Juno_Vec2_F32_Dot) include/juno/math/juno_vec.h:235 (Juno_Vec2_I32_Dot) include/juno/math/juno_vec.h:301 (Juno_Vec3_F64_Dot) include/juno/math/juno_vec.h:372 (Juno_Vec3_F32_Dot) include/juno/math/juno_vec.h:442 (Juno_Vec3_I32_Dot)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:73 (test_vec2_f64_dot) tests/test_math.c:140 (test_vec2_f32_dot) tests/test_math.c:207 (test_vec2_i32_dot) tests/test_math.c:274 (test_vec3_f64_dot) tests/test_math.c:344 (test_vec3_f32_dot) tests/test_math.c:414 (test_vec3_i32_dot)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-009: 2D Cross Product	include/juno/math/juno_vec.h:119 (Juno_Vec2_F64_Cross) include/juno/math/juno_vec.h:182 (Juno_Vec2_F32_Cross) include/juno/math/juno_vec.h:244 (Juno_Vec2_I32_Cross)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:83 (test_vec2_f64_cross) tests/test_math.c:150 (test_vec2_f32_cross) tests/test_math.c:217 (test_vec2_i32_cross)	Test	Complete
REQ-MATH-010: 3D Cross Product	include/juno/math/juno_vec.h:311 (Juno_Vec3_F64_Cross) include/juno/math/juno_vec.h:381 (Juno_Vec3_F32_Cross) include/juno/math/juno_vec.h:451 (Juno_Vec3_I32_Cross)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:284 (test_vec3_f64_cross) tests/test_math.c:354 (test_vec3_f32_cross) tests/test_math.c:424 (test_vec3_i32_cross)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-011: Squared L2 Norm	include/juno/math/juno_vec.h:129 (Juno_Vec2_F64_L2Norm2) include/juno/math/juno_vec.h:191 (Juno_Vec2_F32_L2Norm2) include/juno/math/juno_vec.h:254 (Juno_Vec2_I32_L2Norm2) include/juno/math/juno_vec.h:325 (Juno_Vec3_F64_L2Norm2) include/juno/math/juno_vec.h:395 (Juno_Vec3_F32_L2Norm2) include/juno/math/juno_vec.h:465 (Juno_Vec3_I32_L2Norm2)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:93 (test_vec2_f64_L2) tests/test_math.c:160 (test_vec2_f32_L2) tests/test_math.c:227 (test_vec2_i32_L2) tests/test_math.c:297 (test_vec3_f64_L2) tests/test_math.c:367 (test_vec3_f32_L2) tests/test_math.c:437 (test_vec3_i32_L2)	Test	Complete
REQ-MATH-012: Quaternion Addition	include/juno/math/juno_vec.h:476 (Juno_RQuat_F64_Add)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:446 (test_rquat_f64_add)	Test	Complete
REQ-MATH-013: Quaternion Subtraction	include/juno/math/juno_vec.h:490 (Juno_RQuat_F64_Sub)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:458 (test_rquat_f64_sub)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MATH-014: Quaternion Scalar Multiplication	include/juno/math/juno_vec.h:503 (Juno_RQuat_F64_Mult)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:470 (test_rquat_f64_mult_scalar)	Test	Complete
REQ-MATH-015: Hamilton Product	include/juno/math/juno_vec.h:523 (Juno_RQuat_F64_HamProd)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:482 (test_rquat_f64_hamprod) tests/test_math.c:495 (test_rquat_f64_hamprod_general)	Test	Complete
REQ-MATH-016: Quaternion Conjugate	include/juno/math/juno_vec.h:539 (Juno_RQuat_F64_Conj)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:509 (test_rquat_f64_conj)	Test	Complete
REQ-MATH-017: Quaternion Reciprocal	include/juno/math/juno_vec.h:562 (Juno_RQuat_F64_Recip)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:538 (test_rquat_f64_recip)	Test	Complete
REQ-MATH-018: Quaternion Squared Norm	include/juno/math/juno_vec.h:551 (Juno_RQuat_F64_L2Norm2)	docs/sdd/modules/math.adoc:1 (Math Module)	tests/test_math.c:520 (test_rquat_f64_l2norm2) tests/test_math.c:529 (test_rquat_f64_l2norm)	Test	Complete

3.10. Module: MEMORY

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MEMORY-001: Memory Allocator Interface	include/juno/memory/memory_api.h:80 (<unknown>)	docs/sdd/modules/memory.adoc:1 (Memory Module)	*none*	Inspection	Complete
REQ-MEMORY-002: Allocator Pointer API Dependency	include/juno/memory/memory_api.h:72 (JUNO_MODULE_ROOT)	docs/sdd/modules/memory.adoc:1 (Memory Module)	*none*	Inspection	Complete
REQ-MEMORY-003: Block Allocator Initialization	src/juno_memory_block.c:248 (JunoMemory_BlockInit)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:272 (test_invalid_init_parameters) tests/test_memory.c:428 (test_bad_api)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MEMORY-004: Block Allocator Get	src/juno_memory_block.c:65 (Juno_MemoryBlkGet)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:83 (test_nominal_single_alloc_and_free) tests/test_memory.c:123 (test_nominal_multiple_alloc_and_free) tests/test_memory.c:185 (test_negative_memory_empty) tests/test_memory.c:234 (test_negative_memory_full) tests/test_memory.c:373 (test_generic_memory_get_put) tests/test_memory.c:406 (test_zero_size_allocation)	Test	Complete
REQ-MEMORY-005: Block Allocator Free Block Reuse	src/juno_memory_block.c:65 (Juno_MemoryBlkGet)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:123 (test_nominal_multiple_alloc_and_free)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MEMORY-006: Block Allocator Put	src/juno_memory_block.c:158 (Juno_MemoryBlkPut)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:83 (test_nominal_single_alloc_and_free) tests/test_memory.c:123 (test_nominal_multiple_alloc_and_free) tests/test_memory.c:373 (test_generic_memory_get_put)	Test	Complete
REQ-MEMORY-007: Block Allocator Double Free Rejection	src/juno_memory_block.c:158 (Juno_MemoryBlkPut)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:290 (test_double_free)	Test	Complete
REQ-MEMORY-008: Block Allocator Invalid Address Rejection	src/juno_memory_block.c:158 (Juno_MemoryBlkPut)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:319 (test_free_unallocated) tests/test_memory.c:464 (test_invalid_size_and_addr)	Test	Complete
REQ-MEMORY-009: Block Allocator Update	src/juno_memory_block.c:139 (Juno_MemoryBlkUpdate)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:341 (test_update_memory)	Test	Complete
REQ-MEMORY-010: Block Allocator No Memory Move	src/juno_memory_block.c:139 (Juno_MemoryBlkUpdate)	docs/sdd/modules/memory.adoc:1 (Memory Module)	tests/test_memory.c:341 (test_update_memory)	Test	Complete

3.11. Module: MODULE

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MODULE-001: Module Root Structure	include/juno/module.h:128 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System)	*none*	Inspection	Complete
REQ-MODULE-002: Module Root Failure Handler	include/juno/module.h:128 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System)	*none*	Inspection	Complete
REQ-MODULE-003: Module Derivation	include/juno/module.h:160 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System)	*none*	Inspection	Complete
REQ-MODULE-004: Module Union	include/juno/module.h:111 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System)	*none*	Inspection	Complete
REQ-MODULE-005: Result Type	include/juno/module.h:197 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System) docs/sdd/modules/types.adoc:1 (Common Types Module)	*none*	Inspection	Complete
REQ-MODULE-006: Option Type	include/juno/module.h:241 (<unknown>)	docs/sdd/modules/module_system.adoc:1 (Module System)	*none*	Inspection	Complete

3.12. Module: MTX

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-MTX-001: Mutex Root Structure	include/juno/mp/mtx_api.h:44 (JUNO_MODULE_ROOT)	docs/sdd/modules/mutex.adoc:1 (Mutex Module)	*none*	Inspection	Complete
REQ-MTX-002: TryLock Operation	include/juno/mp/mtx_api.h:50 (JUNO_RESULT_BOOL_T)	docs/sdd/modules/mutex.adoc:1 (Mutex Module)	*none*	Inspection	Complete
REQ-MTX-003: Lock Operation	include/juno/mp/mtx_api.h:53 (JUNO_STATUS_T)	docs/sdd/modules/mutex.adoc:1 (Mutex Module)	*none*	Inspection	Complete
REQ-MTX-004: Free Operation	include/juno/mp/mtx_api.h:56 (JUNO_STATUS_T)	docs/sdd/modules/mutex.adoc:1 (Mutex Module)	*none*	Inspection	Complete
REQ-MTX-005: Mutex Vtable Validation	include/juno/mp/mtx_api.h:61 (JunoMp_MtxVerify)	docs/sdd/modules/mutex.adoc:1 (Mutex Module)	tests/test_mtx.c:77 (test_mtx_verify_valid) tests/test_mtx.c:84 (test_mtx_verify_null_root) tests/test_mtx.c:91 (test_mtx_verify_null_api) tests/test_mtx.c:99 (test_mtx_verify_null_trylock)	Test	Complete

3.13. Module: POINTER

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-POINTER-001: Pointer Descriptor Structure	include/juno/memory/pointer_api.h:55 (JUNO_TRAIT_ROOT)	docs/sdd/modules/pointer.adoc:1 (Pointer Module)	*none*	Inspection	Complete
REQ-POINTER-002: Pointer Copy	include/juno/memory/pointer_api.h:156 (JunoMemory_PointerApiVerify)	docs/sdd/modules/pointer.adoc:1 (Pointer Module)	tests/test_map.c:833 (test_map_entry_copy_nominal) tests/test_map.c:849 (test_map_entry_copy_null_dest) tests/test_pointer.c:89 (test_pointer_copy_success) tests/test_pointer.c:100 (test_pointer_copy_null_src_addr) tests/test_stack.c:1032 (test_stack_pointer_copy_invalid_dest) tests/test_stack.c:1043 (test_stack_pointer_copy_invalid_src)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-POINTER-003: Pointer Reset	include/juno/memory/pointer_api.h:156 (JunoMemory_PointerApiVerify)	docs/sdd/modules/pointer.adoc:1 (Pointer Module)	tests/test_map.c:860 (test_map_entry_reset_nominal) tests/test_pointer.c:113 (test_pointer_reset_success) tests/test_pointer.c:123 (test_pointer_reset_null_addr) tests/test_stack.c:1054 (test_stack_pointer_reset_invalid) tests/test_stack.c:1062 (test_stack_pointer_reset_valid)	Test	Complete
REQ-POINTER-004: Pointer Descriptor Validation	include/juno/memory/pointer_api.h:188 (JunoMemory_PointerVerify)	docs/sdd/modules/pointer.adoc:1 (Pointer Module)	tests/test_pointer.c:135 (test_pointer_verify_valid) tests/test_pointer.c:143 (test_pointer_verify_null_api) tests/test_pointer.c:151 (test_pointer_verify_null_addr) tests/test_pointer.c:159 (test_pointer_verify_zero_size)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-POINTER-005: Pointer Type Verification	include/juno/memory/pointer_api.h:129 (JunoMemory_PointerVerify)	docs/sdd/modules/pointer.adoc:1 (Pointer Module)	tests/test_pointer.c:172 (test_pointer_verify_type_match) tests/test_pointer.c:180 (test_pointer_verify_type_wrong_size) tests/test_pointer.c:189 (test_pointer_verify_type_wrong_alignment)	Test	Complete

3.14. Module: QUEUE

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-001: Queue Initialization	src/juno_buff_queue.c:106 (JunoDs_QueueInit) include/juno/ds/queue_api.h:116 (JunoDs_QueueVerify)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:209 (test_queue_init_nominal) tests/test_queue.c:220 (test_queue_init_null_queue) tests/test_queue.c:230 (test_queue_init_null_array) tests/test_queue.c:237 (test_queue_init_zero_capacity) tests/test_queue.c:729 (test_queue_verify_valid_queue) tests/test_queue.c:739 (test_queue_verify_null_queue) tests/test_queue.c:746 (test_queue_verify_invalid_api) tests/test_queue.c:759 (test_queue_api_verify_valid) tests/test_queue.c:769 (test_queue_api_verify_null) tests/test_queue.c:776 (test_queue_api_verify_missing_functions)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-002: Queue Initial State	src/juno_buff_queue.c:106 (JunoDs_QueueInit)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:209 (test_queue_init_nominal)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-003: Queue Enqueue	src/juno_buff_queue.c:24 (JunoDs_QueuePush)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:249 (test_queue_push_single_item) tests/test_queue.c:266 (test_queue_push_multiple_items) tests/test_queue.c:290 (test_queue_push_to_full_capacity) tests/test_queue.c:331 (test_queue_push_null_queue) tests/test_queue.c:341 (test_queue_push_invalid_pointer) tests/test_queue.c:554 (test_queue_circular_wraparound) tests/test_queue.c:605 (test_queue_wraparound_at_boundary) tests/test_queue.c:642 (test_queue_stress_push_pop_cycles) tests/test_queue.c:673 (test_queue_interleaved_operations) tests/test_queue.c:803 (test_queue_single_element_capacity) tests/test_queue.c:831 (test_queue_boundary_index_calculation) tests/test_queue.c:860	Test	Complete
			(test_queue_data_int		

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-004: Queue Full Rejection	src/juno_buff_queue.c:24 (JunoDs_QueuePush)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:308 (test_queue_push_overflow)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-005: Queue Dequeue	src/juno_buff_queue.c:50 (JunoDs_QueuePop)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:356 (test_queue_pop_single_item) tests/test_queue.c:379 (test_queue_pop_multiple_items_fifo_order) tests/test_queue.c:421 (test_queue_pop_null_queue) tests/test_queue.c:430 (test_queue_pop_invalid_pointer) tests/test_queue.c:554 (test_queue_circular_wraparound) tests/test_queue.c:605 (test_queue_wraparound_at_boundary) tests/test_queue.c:642 (test_queue_stress_push_pop_cycles) tests/test_queue.c:673 (test_queue_interleaved_operations) tests/test_queue.c:803 (test_queue_single_element_capacity) tests/test_queue.c:860 (test_queue_data_integrity_after_operations)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-QUEUE-006: Queue Empty Dequeue Rejection	src/juno_buff_queue.c:50 (JunoDs_QueuePop)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:409 (test_queue_pop_empty_queue)	Test	Complete
REQ-QUEUE-007: Queue Peek	src/juno_buff_queue.c:78 (JunoDs_QueuePeek)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:452 (test_queue_peek_single_item) tests/test_queue.c:478 (test_queue_peek_multiple_times) tests/test_queue.c:515 (test_queue_peek_null_queue) tests/test_queue.c:522 (test_queue_peek_after_pop)	Test	Complete
REQ-QUEUE-008: Queue Empty Peek Rejection	src/juno_buff_queue.c:78 (JunoDs_QueuePeek)	docs/sdd/modules/queue.adoc:1 (Queue Module)	tests/test_queue.c:505 (test_queue_peek_empty_queue)	Test	Complete

3.15. Module: SB

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SB-001: Broker Root Structure	include/juno/sb/broker_api.h:79 (JUNO_MODULE_ROOT)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SB-002: Broker API Interface	include/juno/sb/broker_api.h:89 (<unknown>)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	*none*	Inspection	Complete
REQ-SB-003: Broker Initialization	src/juno_broker.c:35 (JunoSb_BrokerInit)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:190 (test_sb_broker_init_success) tests/test_sb.c:203 (test_sb_broker_init_null_broker) tests/test_sb.c:212 (test_sb_broker_init_null_registry)	Test	Complete
REQ-SB-004: Pipe Derivation	include/juno/sb/broker_api.h:72 (JUNO_MODULE_DERIVE)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	*none*	Inspection	Complete
REQ-SB-005: Pipe Initialization	include/juno/sb/broker_api.h:138 (JunoSb_PipeInit)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:225 (test_sb_pipe_init_success) tests/test_sb.c:234 (test_sb_pipe_init_null_pipe)	Test	Complete
REQ-SB-006: Publish Fan-Out	src/juno_broker.c:50 (Publish)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:280 (test_sb_publish_fan_out)	Test	Complete
REQ-SB-007: Publish Non-Matching Skip	src/juno_broker.c:50 (Publish)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:309 (test_sb_publish_skip_non_matching)	Test	Complete
REQ-SB-008: Register Subscriber	src/juno_broker.c:75 (RegisterSubscriber)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:246 (test_sb_register_subscriber)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SB-009: Registry Full Rejection	src/juno_broker.c:75 (RegisterSubscriber)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:259 (test_sb_register_full_rejection)	Test	Complete
REQ-SB-010: Broker Vtable Validation	include/juno/sb/broker_api.h:113 (JunoSb_BrokerVerify)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:336 (test_sb_broker_verify_null_api)	Test	Complete
REQ-SB-011: Broker Registry Validation	include/juno/sb/broker_api.h:113 (JunoSb_BrokerVerify)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:346 (test_sb_broker_verify_null_registry) tests/test_sb.c:356 (test_sb_broker_verify_zero_capacity)	Test	Complete
REQ-SB-012: Pipe Validation	include/juno/sb/broker_api.h:125 (JunoSb_PipeVerify)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	tests/test_sb.c:366 (test_sb_pipe_verify_valid) tests/test_sb.c:374 (test_sb_pipe_verify_null)	Test	Complete
REQ-SB-013: Single Thread Scope	include/juno/sb/broker_api.h:50 (<unknown>)	docs/sdd/modules/software_bus.adoc:1 (Software Bus (Broker) Module)	*none*	Inspection	Complete

3.16. Module: SCH

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SCH-001: Scheduler Root Structure	include/juno/sch/juno_sch_api.h:61 (JUNO_MODULE_ROOT)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SCH-002: Scheduler API Interface	include/juno/sch/juno_sch_api.h:70 (<unknown>)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	*none*	Inspection	Complete
REQ-SCH-003: Schedule Table Layout	include/juno/sch/juno_sch_api.h:56 (<unknown>)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	*none*	Inspection	Complete
REQ-SCH-004: Execute Major Frame	include/juno/sch/juno_sch_api.h:74 (JUNO_STATUS_T)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	tests/test_sch.c:153 (test_sch_execute_invokes_all_apps) tests/test_sch.c:162 (test_sch_execute_skips_null_slot)	Test	Complete
REQ-SCH-005: Minor Frame Period Query	include/juno/sch/juno_sch_api.h:77 (JUNO_TIMESTAMP_RESULT_T)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	tests/test_sch.c:177 (test_sch_get_minor_frame_period)	Test	Complete
REQ-SCH-006: Major Frame Period Computation	include/juno/sch/juno_sch_api.h:80 (JUNO_TIMESTAMP_RESULT_T)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	tests/test_sch.c:190 (test_sch_get_major_frame_period)	Test	Complete
REQ-SCH-007: Time Source Dependency	include/juno/sch/juno_sch_api.h:61 (JUNO_MODULE_ROOT)	docs/sdd/modules/scheduler.adoc:1 (Scheduler Module)	*none*	Inspection	Complete

3.17. Module: SM

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SM-001: SM Root Structure	include/juno/sm/sm_api.h:76 (JUNO_MODULE_ROOT)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	*none*	Inspection	Complete
REQ-SM-002: State Root Structure	include/juno/sm/sm_api.h:59 (JUNO_MODULE_ROOT)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	*none*	Inspection	Complete
REQ-SM-003: State API Interface	include/juno/sm/sm_api.h:65 (JUNO_STATUS_T)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	*none*	Inspection	Complete
REQ-SM-004: SM Initialization	include/juno/sm/sm_api.h:126 (JunoSm_Init)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:91 (test_sm_init_success) tests/test_sm.c:103 (test_sm_init_null_root) tests/test_sm.c:110 (test_sm_init_null_start_state)	Test	Complete
REQ-SM-005: State Initialization	include/juno/sm/sm_api.h:104 (JunoSm_StateInit)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:121 (test_sm_state_init_with_next) tests/test_sm.c:136 (test_sm_state_init_without_next) tests/test_sm.c:148 (test_sm_state_init_null_state) tests/test_sm.c:156 (test_sm_state_init_null_api)	Test	Complete
REQ-SM-006: Get Current State	include/juno/sm/sm_api.h:139 (JunoSm_GetCurrentState)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:168 (test_sm_get_current_state) tests/test_sm.c:179 (test_sm_get_current_state_null_root)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SM-007: State Transition	include/juno/sm/sm_api.h:151 (JunoSm_TransitionState)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:190 (test_sm_transition_to_next) tests/test_sm.c:220 (test_sm_transition_null_root)	Test	Complete
REQ-SM-008: No Transition When None	include/juno/sm/sm_api.h:151 (JunoSm_TransitionState)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:207 (test_sm_transition_stays_when_none)	Test	Complete
REQ-SM-009: SM Verification	include/juno/sm/sm_api.h:83 (JunoSm_Verify)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:231 (test_sm_verify_valid) tests/test_sm.c:241 (test_sm_verify_null_root) tests/test_sm.c:248 (test_sm_verify_null_current_state)	Test	Complete
REQ-SM-010: State Verification	include/juno/sm/sm_api.h:91 (JunoSm_StateVerify)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:260 (test_sm_state_verify_valid) tests/test_sm.c:270 (test_sm_state_verify_null_state) tests/test_sm.c:277 (test_sm_state_verify_null_api) tests/test_sm.c:286 (test_sm_state_verify_null_sm)	Test	Complete
REQ-SM-011: Next State Pointer Verification	include/juno/sm/sm_api.h:91 (JunoSm_StateVerify)	docs/sdd/modules/state_machine.adoc:1 (State Machine Module)	tests/test_sm.c:295 (test_sm_state_verify_next_null_when_some) tests/test_sm.c:306 (test_sm_state_verify_next_valid_when_some)	Test	Complete

3.18. Module: STACK

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STACK-001: Stack Initialization	src/juno_buff_stack.c:101 (JunoDs_StackInit)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:211 (test_stack_init_nominal) tests/test_stack.c:221 (test_stack_init_null_stack) tests/test_stack.c:228 (test_stack_init_null_api) tests/test_stack.c:236 (test_stack_init_zero_capacity) tests/test_stack.c:705 (test_stack_verify_valid_stack) tests/test_stack.c:715 (test_stack_verify_null_stack) tests/test_stack.c:722 (test_stack_verify_invalid_api) tests/test_stack.c:735 (test_stack_verify_missing_push_function) tests/test_stack.c:751 (test_stack_verify_missing_pop_function) tests/test_stack.c:767 (test_stack_verify_missing_peek_function) tests/test_stack.c:1170 (test_stack_init_with_array_verify_failure)	Test	Complete
REQ-STACK-002: Stack Initial State	src/juno_buff_stack.c:101 (JunoDs_StackInit)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:211 (test_stack_init_nominal)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STACK-003: Stack Push	src/juno_buff_stack.c:27 (JunoDs_StackPush)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:248 (test_stack_push_single_item) tests/test_stack.c:265 (test_stack_push_multiple_items) tests/test_stack.c:289 (test_stack_push_to_full_capacity) tests/test_stack.c:330 (test_stack_push_null_stack) tests/test_stack.c:340 (test_stack_push_invalid_pointer) tests/test_stack.c:578 (test_stack_stress_push_pop_cycles) tests/test_stack.c:609 (test_stack_interleaved_operations) tests/test_stack.c:667 (test_stack_fill_empty_cycles) tests/test_stack.c:787 (test_stack_single_element_capacity) tests/test_stack.c:820 (test_stack_data_integrity_after_operations) tests/test_stack.c:848 (test_stack_alternating_push_pop) tests/test_stack.c:966 (test_stack_push_at_exact_capacity) tests/test_stack.c:1079 (test_stack_push_with_setAt_failure)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STACK-004: Stack Full Rejection	src/juno_buff_stack.c:27 (JunoDs_StackPush)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:307 (test_stack_push_overflow) tests/test_stack.c:966 (test_stack_push_at_exact_capacity)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STACK-005: Stack Pop	src/juno_buff_stack.c:52 (JunoDs_StackPop)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:355 (test_stack_pop_single_item) tests/test_stack.c:378 (test_stack_pop_multiple_items_lifo_order) tests/test_stack.c:420 (test_stack_pop_null_stack) tests/test_stack.c:429 (test_stack_pop_invalid_pointer) tests/test_stack.c:578 (test_stack_stress_push_pop_cycles) tests/test_stack.c:609 (test_stack_interleaved_operations) tests/test_stack.c:667 (test_stack_fill_empty_cycles) tests/test_stack.c:787 (test_stack_single_element_capacity) tests/test_stack.c:820 (test_stack_data_integrity_after_operations) tests/test_stack.c:848 (test_stack_alternating_push_pop) tests/test_stack.c:992 (test_stack_pop_until_empty) tests/test_stack.c:1099 (test_stack_pop_with_getAt_failure) tests/test_stack.c:1148 (test_stack_pop_copy_failure)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STACK-006: Stack Empty Pop Rejection	src/juno_buff_stack.c:52 (JunoDs_StackPop)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:408 (test_stack_pop_empty_stack) tests/test_stack.c:992 (test_stack_pop_until_empty)	Test	Complete
REQ-STACK-007: Stack Peek	src/juno_buff_stack.c:77 (JunoDs_StackPeek)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:451 (test_stack_peek_single_item) tests/test_stack.c:477 (test_stack_peek_top_element) tests/test_stack.c:501 (test_stack_peek_multiple_times) tests/test_stack.c:538 (test_stack_peek_null_stack) tests/test_stack.c:545 (test_stack_peek_after_pop) tests/test_stack.c:880 (test_stack_peek_consistency_during_operations) tests/test_stack.c:1125 (test_stack_peek_with_getAt_failure)	Test	Complete
REQ-STACK-008: Stack Empty Peek Rejection	src/juno_buff_stack.c:77 (JunoDs_StackPeek)	docs/sdd/modules/stack.adoc:1 (Stack Module)	tests/test_stack.c:528 (test_stack_peek_empty_stack)	Test	Complete

3.19. Module: STATUS

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-STATUS-001: Unified Status Type	include/juno/status.h:50 (<unknown>)	docs/sdd/modules/status.adoc:1 (Status Module)	*none*	Inspection	Complete
REQ-STATUS-002: Canonical Status Codes	include/juno/status.h:57 (<unknown>)	docs/sdd/modules/status.adoc:1 (Status Module)	*none*	Inspection	Complete
REQ-STATUS-003: Failure Handler Callback Type	include/juno/status.h:107 (<unknown>)	docs/sdd/modules/status.adoc:1 (Status Module)	*none*	Inspection	Complete
REQ-STATUS-004: Failure Handler Non-Interference	include/juno/status.h:127 (<unknown>)	docs/sdd/modules/status.adoc:1 (Status Module)	*none*	Inspection	Complete

3.20. Module: SYS

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SYS-001: Zero Dynamic Allocation	CMakeLists.txt:75 (list)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SYS-002: Caller-Owned Memory	include/juno/module.h:54 (JUNO_MODULE_DECLARE)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete
REQ-SYS-003: Freestanding C11 Compatibility	CMakeLists.txt:75 (list)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Analysis	Complete
REQ-SYS-004: Explicit Status Code Returns	include/juno/status.h:40 (<unknown>)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete
REQ-SYS-005: No Silent Failures	include/juno/status.h:40 (<unknown>)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete
REQ-SYS-006: Dependency Injection via Vtables	include/juno/module.h:54 (JUNO_MODULE_DECLARE)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete
REQ-SYS-007: No Global Dependencies	include/juno/macros.h:50 (<unknown>)	docs/sdd/modules/macros.adoc:1 (Macros Module)	*none*	Inspection	Complete
REQ-SYS-008: Precondition Verification	include/juno/macros.h:50 (<unknown>) include/juno/macros.h:88 (<unknown>)	docs/sdd/modules/macros.adoc:1 (Macros Module)	tests/test_pointer.c:151 (test_pointer_verify_null_addr)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-SYS-009: No Global Mutable State	include/juno/module.h:54 (JUNO_MODULE_DECLARE)	docs/sdd/sdd.adoc:58 (System Architecture)	*none*	Inspection	Complete
REQ-SYS-010: API Vtable Validation	include/juno/macros.h:50 (<unknown>)	docs/sdd/modules/macros.adoc:1 (Macros Module)	tests/test_array.c:238 (test_array_verify_null_api)	Test	Complete
REQ-SYS-011: Injected Dependency Validation	include/juno/macros.h:50 (<unknown>)	docs/sdd/modules/macros.adoc:1 (Macros Module)	tests/test_sb.c:346 (test_sb_broker_verify_null_registry)	Test	Complete

3.21. Module: TIME

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-TIME-001: Timestamp Representation	include/juno/time/time_api.h:104 (<unknown>)	docs/sdd/modules/time.adoc:1 (Time Module)	*none*	Inspection	Complete
REQ-TIME-002: Time API Interface	include/juno/time/time_api.h:128 (JUNO_TIMESTAMP_RESULT_T)	docs/sdd/modules/time.adoc:1 (Time Module)	*none*	Inspection	Complete
REQ-TIME-003: AddTime Operation	src/juno_time.c:26 (JunoTime_AddTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:86 (test_AddTime_no_subseconds_overflow)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-TIME-004: AddTime Subsecond Carry-Over	src/juno_time.c:26 (JunoTime_AddTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:97 (test_AddTime_with_subseconds_overflow)	Test	Complete
REQ-TIME-005: SubtractTime Operation	src/juno_time.c:49 (JunoTime_SubtractTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:110 (test_SubtractTime_no_borrow)	Test	Complete
REQ-TIME-006: SubtractTime Subsecond Borrow	src/juno_time.c:49 (JunoTime_SubtractTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:122 (test_SubtractTime_borrow_subseconds)	Test	Complete
REQ-TIME-007: SubtractTime Negative Saturation	src/juno_time.c:49 (JunoTime_SubtractTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:135 (test_SubtractTime_invalid_lower_seconds) tests/test_time.c:146 (test_SubtractTime_invalid_zero_seconds_insufficient_subseconds)	Test	Complete
REQ-TIME-008: SubtractTime Negative Error	src/juno_time.c:49 (JunoTime_SubtractTime)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:135 (test_SubtractTime_invalid_lower_seconds) tests/test_time.c:146 (test_SubtractTime_invalid_zero_seconds_insufficient_subseconds)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-TIME-009: Timestamp to Nanoseconds Conversion	src/juno_time.c:78 (JunoTime_TimestampToNanos)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:157 (test_TimestampToNanos_success_integer) tests/test_time.c:166 (test_TimestampToNanos_success_fractional)	Test	Complete
REQ-TIME-010: Timestamp to Microseconds Conversion	src/juno_time.c:115 (JunoTime_TimestampToMicros)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:177 (test_TimestampToMicros_success_integer) tests/test_time.c:186 (test_TimestampToMicros_success_fractional)	Test	Complete
REQ-TIME-011: Timestamp to Milliseconds Conversion	src/juno_time.c:149 (JunoTime_TimestampToMillis)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:196 (test_TimestampToMillis_success_integer) tests/test_time.c:205 (test_TimestampToMillis_success_fractional)	Test	Complete
REQ-TIME-012: Nanoseconds to Timestamp Conversion	src/juno_time.c:184 (JunoTime_NanosToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:215 (test_NanosToTimestamp_zero) tests/test_time.c:225 (test_NanosToTimestamp_integer_and_fractional)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-TIME-013: Microseconds to Timestamp Conversion	src/juno_time.c:202 (JunoTime_MicrosToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:240 (test_MicrosToTimestamp_zero) tests/test_time.c:250 (test_MicrosToTimestamp_integer_and_fractional)	Test	Complete
REQ-TIME-014: Milliseconds to Timestamp Conversion	src/juno_time.c:220 (JunoTime_MillisToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:265 (test_MillisToTimestamp_zero) tests/test_time.c:275 (test_MillisToTimestamp_integer_and_fractional)	Test	Complete
REQ-TIME-015: Timestamp to Double Conversion	src/juno_time.c:239 (JunoTime_TimestampToDouble)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:314 (test_timestamp_to_double)	Test	Complete
REQ-TIME-016: Double to Timestamp Conversion	src/juno_time.c:254 (JunoTime_DoubleToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:324 (test_double_to_timestamp_success) tests/test_time.c:333 (test_double_to_timestamp_zero)	Test	Complete
REQ-TIME-017: Double to Timestamp Range Rejection	src/juno_time.c:254 (JunoTime_DoubleToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:342 (test_double_to_timestamp_negative_rejected)	Test	Complete

Requirement	Code Location	Design Location	Test Location	Verification Method	Status
REQ-TIME-018: Conversion Overflow Detection	src/juno_time.c:78 (JunoTime_TimestampToNanos) src/juno_time.c:115 (JunoTime_TimestampToMicros) src/juno_time.c:149 (JunoTime_TimestampToMillis)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:349 (test_timestamp_to_nanos_overflow)	Test	Complete
REQ-TIME-019: Null Pointer Rejection	src/juno_time.c:78 (JunoTime_TimestampToNanos) src/juno_time.c:115 (JunoTime_TimestampToMicros) src/juno_time.c:149 (JunoTime_TimestampToMillis) src/juno_time.c:184 (JunoTime_NanosToTimestamp) src/juno_time.c:202 (JunoTime_MicrosToTimestamp) src/juno_time.c:220 (JunoTime_MillisToTimestamp) src/juno_time.c:239 (JunoTime_TimestampToDouble) src/juno_time.c:254 (JunoTime_DoubleToTimestamp)	docs/sdd/modules/time.adoc:1 (Time Module)	tests/test_time.c:362 (test_double_to_timestamp_null_root)	Test	Complete